

PROJECT PRESENTATION

Society Management Portal

A highly scalable, distributed platform designed to digitize and automate residential complex operations.

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Abstract

The Society/Apartment Management Portal is a web-based application designed to **simplify and digitize** the day-to-day management activities of residential apartments.

Currently, activities such as maintenance fee collection, complaint handling, visitor entry, and amenity booking are managed manually or through disjointed systems.

Our proposed system provides a **centralized platform** utilizing a structured database to securely maintain resident, payment, complaint, visitor, and booking information in one cohesive ecosystem.



Problem Statement



Inefficient Records

Manual handling of maintenance fees, complaints, and visitors leads to lost data and severe administrative bottlenecks.



Disconnected Comms

The absence of a unified system causes delayed updates and miscommunication between property administrators and residents.



Difficult Tracking

Relying on physical ledgers or fragmented Excel sheets makes it nearly impossible to accurately track financial dues and payments.



Compromised Security

Traditional paper-based visitor logbooks are easily tampered with, slow down gate operations, and lack searchability.

Proposed Solution



Centralized Digital Platform

A unified portal where administrators and residents can efficiently manage all residential activities.



Financial Transparency

Automated tracking of maintenance bills with an integrated online payment facility for residents.



Digital Gate-Pass

Modernized visitor management system for maintaining accurate records and improving entry security.



Structured Architecture

A robust backend database to securely link resident profiles, payments, complaints, and bookings.



System Architecture



Client Application

React.js frontend providing distinct dashboards for Admins, Residents, and Gate Security.



API Gateway & Logic

FastAPI (Python) microservices handling auth, routing, and core business logic asynchronously.



Data Persistence

MongoDB via Motor (Async driver) ensuring non-blocking, scalable document storage.

Core Modules



Resident & Flat Directory

Maintains centralized, structured information for all apartments. Serves as the single source of truth for the property database.



Billing & Payments

Automated maintenance fee generation and seamless online collection.



Complaint Handling

Ticket-based issue tracking allowing residents to raise and monitor maintenance requests.



Visitor Gate-Pass

Digital entry logs, pre-approvals, and real-time verification at the security gate.



Amenity Booking

Conflict-free, scheduled access to shared facilities like the gym, pool, or clubhouse.

Technology Stack



React.js

Frontend Framework



FastAPI

Backend Framework



MongoDB

NoSQL Database



Motor Async

Async DB Driver